

Algebra II Triangle Trigonometry Cheatsheet

Module 10

Right Triangle Trig

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

Reciprocal functions:

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}$$

Special Right Triangles

$$45^\circ - 45^\circ - 90^\circ : \quad x, x, x\sqrt{2}$$

$$30^\circ - 60^\circ - 90^\circ : \quad x, x\sqrt{3}, 2x$$

Triangle Labels

Side a is opposite angle A , side b is opposite angle B , and side c is opposite angle C .

Law of Sines

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Use for ASA, AAS, and sometimes SSA.

SSA Ambiguous Case

Given A , a , and b , compute:

$$h = b \sin A$$

$$a < h \quad \text{no triangle}$$

$$a = h \quad \text{one right triangle}$$

$$h < a < b \quad \text{two triangles}$$

$$a \geq b \quad \text{one triangle}$$

Law of Cosines

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

Use for SAS and SSS.

Area

Two sides and included angle:

$$K = \frac{1}{2}ab \sin C$$

Heron's Formula:

$$s = \frac{a + b + c}{2}, \quad K = \sqrt{s(s - a)(s - b)(s - c)}$$

Bearings

True bearings are measured clockwise from north.

$$060^\circ = 60^\circ \text{ clockwise from north}$$

$$\text{N } 35^\circ \text{ E} = 35^\circ \text{ east of north}$$

For bearing word problems, draw north-south-east-west first, mark the given directions, then identify the included triangle angle.

Quick Checks

- Law of Cosines handles the included angle or all three sides.
- Law of Sines can create two possible triangles in SSA.
- Area formula $\frac{1}{2}ab \sin C$ needs the included angle.
- Keep angle units consistent.